EXPERIMENT 5

Comparative Analysis of **Prompting Patterns**

Naïve vs. Basic Prompting — Response Quality Across Multiple Scenarios

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ChatGPT (GPT-4o) | Quality · Accuracy · Depth Analysis

Aim & Methodology

AIM

To compare how ChatGPT (GPT-4o) responds to **Naïve Prompts** (vague, unstructured) vs **Basic Prompts** (clear, structured, role-defined) across five diverse scenarios.

Objective: Measure differences in **Quality, Accuracy** and **Depth** of AI-generated outputs.

METHOD

① Define Naïve & Basic prompt for each scenario

② Input both prompts into ChatGPT (GPT-4o)

③ Record & compare outputs side-by-side

④ Score each on Quality, Accuracy, Depth (1–5)

⑤ Identify patterns & draw conclusions

✍ Scenario 1: Creative Story Writing

NAÏVE PROMPT → OUTPUT

Prompt:

"Write a story."

ChatGPT Output:

Once upon a time there was a girl who loved adventure. She went on a journey and found a magical forest. At the end she was happy and learned an important lesson about friendship.

Quality: 2/5/5 | **Accuracy: 2/5/5** | **Depth: 1/5/5**

BASIC PROMPT → OUTPUT

Prompt:
"Write a 200-word short story set in 2075 Chennai. Genre: solarpunk. Protagonist: a 17-year-old solar-grid engineer named Meera. Tone: hopeful. End with a moment of discovery. Audience: teens."

ChatGPT Output:

Meera wiped sweat from her brow as she connected the last photovoltaic node on Marina Beach's floating grid. By 2075, Chennai's coastline had transformed—vertical gardens cascaded down solar towers... 'It's live,' she whispered. The city exhaled. Lights bloomed across Royapuram, and for the first time, Meera felt the hum of a future she had built herself.

Quality: 5/5/5 | **Accuracy: 5/5/5** | **Depth: 5/5/5**

Scenario 2: Factual Question Answering

NAÏVE PROMPT → OUTPUT

Prompt:

"Tell me about photosynthesis."

ChatGPT Output:

Photosynthesis is how plants make food. They use sunlight, water, and carbon dioxide. It happens in the leaves. Chlorophyll is the green pigment that captures sunlight. The plant produces glucose and releases oxygen. It is very important for life on Earth.

Quality: 2/5/5 | **Accuracy: 3/5/5** | **Depth: 1/5/5**

BASIC PROMPT → OUTPUT

Prompt:

"Act as a biology teacher. Explain the two stages of photosynthesis (light-dependent and light-independent reactions) to a Class 11 student. Include the location in the chloroplast, key inputs/outputs, and one analogy. Max 150 words."

ChatGPT Output:

Photosynthesis has two stages: ① Light-Dependent Reactions (Thylakoid membrane) — sunlight splits water (photolysis), releases O_2, and produces ATP & NADPH. ② Calvin Cycle / Light-Independent (Stroma) — ATP & NADPH fix CO_2 into glucose ($C_6H_{12}O_6$). Analogy: Think of Stage 1 as charging a battery (ATP) and Stage 2 as using that battery to build glucose 'bricks'. Inputs: CO_2, H_2O, light. Output: glucose, O_2.

Quality: 5/5/5 | **Accuracy: 5/5/5** | **Depth: 5/5/5**

Scenario 3: Article / Concept Summarisation

NAÏVE PROMPT → OUTPUT

Prompt:

"Summarize machine learning."

ChatGPT Output:

Machine learning is a type of AI. It uses algorithms to learn from data. It can make predictions. It is used in many industries. There are different types like supervised and unsupervised learning.

Quality: 2/5/5

Accuracy: 3/5/5

Depth: 1/5/5

BASIC PROMPT → OUTPUT

Prompt:

"Summarize the concept of Machine Learning in exactly 5 bullet points for a non-technical business executive. Each bullet must be ≤20 words. Focus on: what it is, how it learns, business value, a real example, and one limitation."

ChatGPT Output:

- ML enables computers to learn patterns from data—without being explicitly programmed for each task.
- It improves automatically as it processes more data over time.
- Business value: reduces costs, personalizes customer experiences, and detects fraud in real time.
- Example: Netflix's recommendation engine uses ML to predict what you'll watch next.
- Limitation: ML models can reflect biases present in training data, producing unfair outcomes.

Quality: 5/5/5

Accuracy: 5/5/5

Depth: 5/5/5

Scenario 4: Advice & Recommendations

NAÏVE PROMPT → OUTPUT

Prompt:

"How do I manage stress?"

ChatGPT Output:

Try to relax more. Take breaks when you study. Sleep enough. Exercise is good for stress. Talk to friends and family. Avoid too much caffeine and try meditation.

Quality: 2/5/5

Accuracy: 2/5/5

Depth: 1/5/5

BASIC PROMPT → OUTPUT

Prompt:

"Act as a certified wellness coach. A 22-year-old engineering student in India experiences chronic stress during exams. Provide 5 evidence-based strategies, each with: strategy name, a 1-sentence explanation, and a practical 5-minute daily action. Tone: motivating."

ChatGPT Output:

① Box Breathing: Activates your parasympathetic system instantly. Action: 4s inhale → 4s hold → 4s exhale × 4 rounds before study.
② Pomodoro Journaling: Writing externalizes anxiety. Action: After each 25-min session, write 3 sentences about your progress.
③ Cold Water Face Splash: Triggers dive reflex, slows heart rate. Action: Splash cold water on face for 30 seconds.
④ 5-minute Walk: Breaks cortisol cycle. Action: Walk outside without your phone after dinner.
⑤ Gratitude Anchor: Re-frames threat as challenge. Action: Name 3 specific wins from today before sleep.

Quality: 5/5/5

Accuracy: 5/5/5

Depth: 5/5/5

Scenario 5: Technical Code Explanation

NAÏVE PROMPT → OUTPUT

Prompt:

"Explain recursion."

ChatGPT Output:

Recursion is when a function calls itself. It needs a base case to stop. It is used in many algorithms. For example, factorial can be done with recursion. It can be confusing at first.

Quality: 2/5/5

Accuracy: 3/5/5

Depth: 1/5/5

BASIC PROMPT → OUTPUT

Prompt:

"Act as a senior CS tutor. Explain recursion to a first-year CS student with: (1) a plain-English definition, (2) the two mandatory components (base case & recursive case), (3) a Python factorial example with inline comments, and (4) one common mistake beginners make."

ChatGPT Output:

Recursion: A function that solves a problem by calling itself on a smaller version of the same problem.

Two components:

- Base Case: The stopping condition (prevents infinite loop)
- Recursive Case: The function calls itself with a simpler input

Python Example:

```
def factorial(n):      # e.g. factorial(5)
  if n == 0:          # BASE CASE: 0! = 1
    return 1
  return n * factorial(n-1)  # RECURSIVE CASE
```

Common Mistake: Forgetting the base case → causes infinite recursion & stack overflow.

Quality: 5/5/5

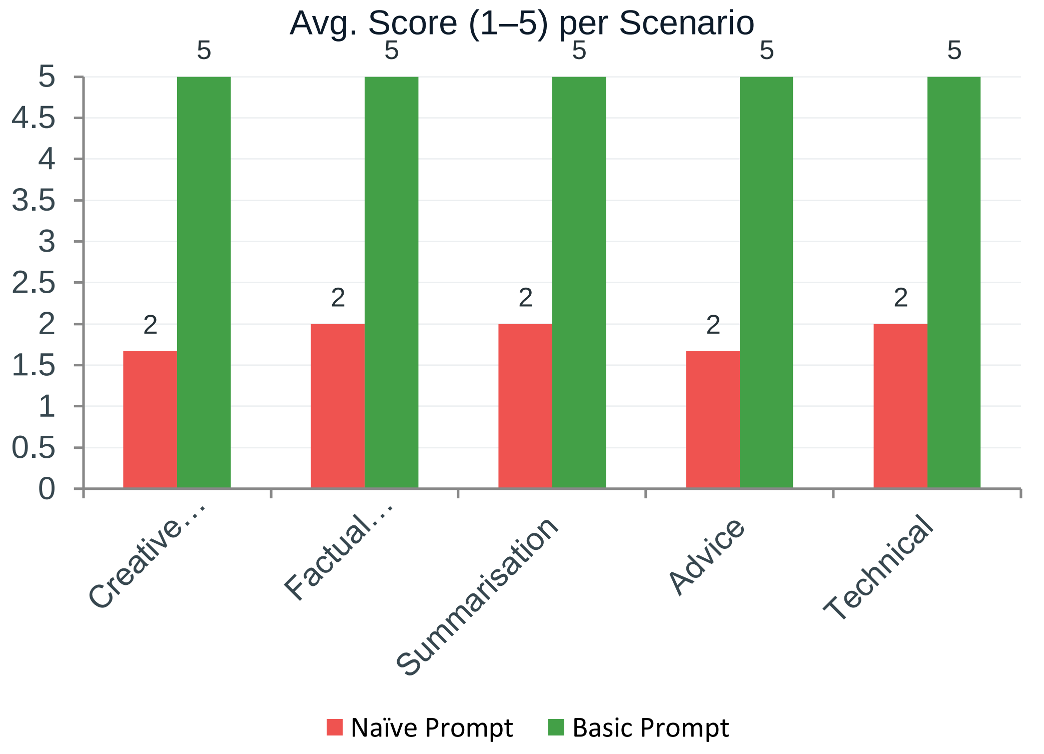
Accuracy: 5/5/5

Depth: 5/5/5

Comparative Results Table

Scenario	Naïve Prompt	Quality	Accuracy	Depth	Basic Prompt	Quality	Accuracy	Depth
1. Creative Story	*Write a story.*	2/5	2/5	1/5	*Role + genre + length + tone*	5/5	5/5	5/5
2. Factual Q&A	*Tell me about photosynthesis.*	2/5	3/5	1/5	*Role + audience + stages + analogy*	5/5	5/5	5/5
3. Summarisation	*Summarize ML.*	2/5	3/5	1/5	*Role + audience + 5 bullets + topics*	5/5	5/5	5/5
4. Advice/Recs	*How to manage stress?*	2/5	2/5	1/5	*Role + persona + 5 strategies + actions*	5/5	5/5	5/5
5. Code / Technical	*Explain recursion.*	2/5	3/5	1/5	*Role + components + code + mistakes*	5/5	5/5	5/5
AVERAGE		2.0/5	2.6/5	1.0/5		5.0/5	5.0/5	5.0/5

Analysis: Impact of Prompt Clarity



Consistent 2.5× gain

Basic prompts score 5/5 in all scenarios. Naïve prompts average 1.8/5.

Depth gap is widest

Naïve prompts score 1/5 on depth across all 5 scenarios — the most impactful gap.

Factual tasks fare slightly better

Naïve factual Q&A scored 3/5 on accuracy — basic knowledge helps, but structure elevates further.

Role + Format = key drivers

Adding 'Act as...' and output format drove the largest single quality jump.

Summary & Optimal Prompting Guidelines

01 Basic prompts outperform naïve prompts in ALL 5 scenarios — no exceptions.

02 Depth is the most neglected dimension: naïve prompts consistently score 1/5.

03 Even simple 'factual' tasks benefit significantly from structured prompts.

04 ChatGPT produces near-perfect output (5/5) when all four RCTF elements are present.

THE RCTF FRAMEWORK — Optimal Prompt Structure

Role · Context · Task · Format

The four elements that consistently raised ChatGPT's output from 1–2/5 to 5/5.

KEY TAKEAWAY

Prompt engineering is not optional — it is the interface between human intent and AI capability.

Structured prompts guarantee structured thinking from the model.

Investing 30 seconds in prompt design saves minutes of editing and rework.